P1

$$\vec{E} = \vec{a} \times \vec{r} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ a_x & a_y & a_z \\ x & y & z \end{vmatrix} = \hat{x}\overbrace{(a_y z - a_z y)}^{E_x} + \hat{y}\overbrace{(a_z x - a_x z)}^{E_y} + \hat{z}\underbrace{(a_x y - a_y x)}_{E_z}$$

Electric field must satisfy $\vec{\nabla} \times \vec{E} = 0$.

check this

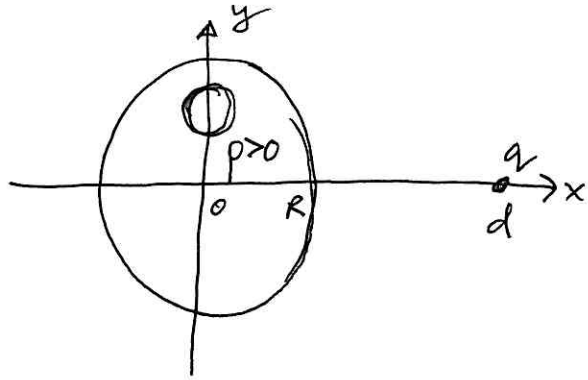
$$\vec{\nabla} \times \vec{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = \hat{x}\left(\frac{\partial}{\partial y} E_z - \frac{\partial E_y}{\partial z}\right) - \hat{y}\left(\frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z}\right) + \hat{z}\left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}\right) =$$

$$= \hat{x}\left(a_x - (-a_x)\right) - \hat{y}\left(-a_y - a_y\right) + \hat{z}\left(a_z - (-a_z)\right) =$$

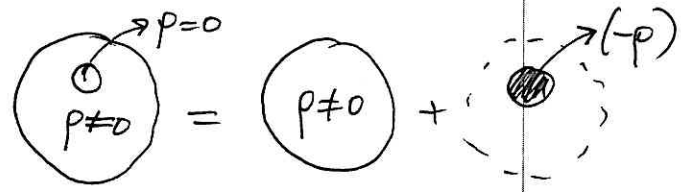
$$= 2a_x \hat{x} + 2a_y \hat{y} + 2a_z \hat{z} = 2\vec{a}.$$

Thus $\vec{\nabla} \times \vec{E} = 2\vec{a} \neq 0 \Rightarrow$ such electrostatic field is not possible.

P2



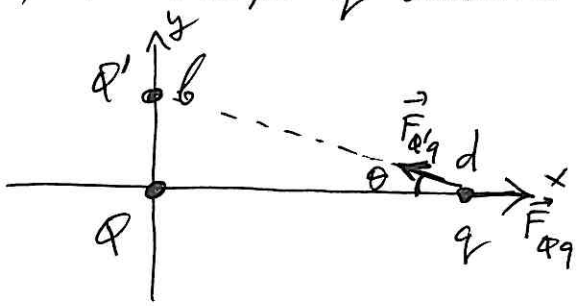
Superposition:



Charged sphere with hollow = uniform sphere of radius R
+ negatively charged sphere of radius a.

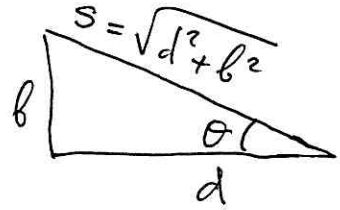
(a) For charge q outside the sphere: $Q = \frac{4\pi}{3}R^3(\rho) > 0$

$Q' = \frac{4\pi}{3}a^3(-\rho) < 0$



$$\vec{F}_q = \vec{F}_{Qq} + \vec{F}_{Q'q}$$

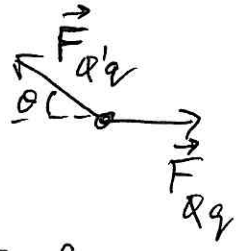
$$\vec{F}_{Qq} = \frac{qQ}{4\pi\varepsilon_0 d^2}\hat{x} \; ; \; \vec{F}_{Q'q} = \frac{+q(Q')}{4\pi\varepsilon_0(d^2+b^2)}(-\cos\theta\,\hat{x} + \sin\theta\,\hat{y})$$



$$\cos\theta = \frac{d}{\sqrt{d^2+b^2}}$$

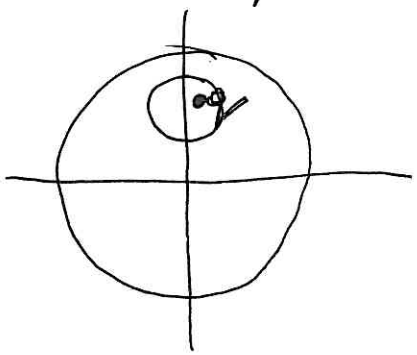
$$\sin\theta = \frac{b}{\sqrt{d^2+b^2}}$$

$$\vec{F}_q = \frac{q}{4\pi\varepsilon_0}\left\{\left(\frac{Q}{d^2} + \frac{Q'}{d^2+b^2}\frac{d}{\sqrt{d^2+b^2}}\right)\hat{x} - \frac{qQ'}{d^2+b^2}\frac{b}{\sqrt{d^2+b^2}}\hat{y}\right\}$$



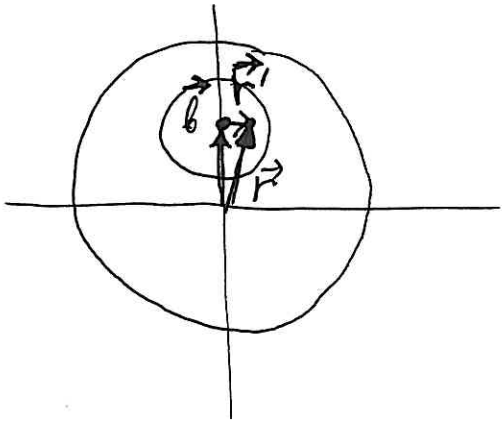
OR $$\vec{F}_q = \frac{q}{4\pi\varepsilon_0}\frac{\frac{4\pi}{3}R^3\rho}{d^2}\left\{\left(1 - \frac{a^3}{R^3}\frac{d^3}{(d^2+b^2)^{3/2}}\right)\hat{x} + \frac{a^3}{R^3}\frac{bd^2}{(b^2+d^2)^{3/2}}\hat{y}\right\}.$$

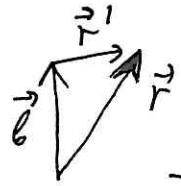
(b) Now charge q is inside the hollow



$$\vec{F}_q = q\,\vec{E}_{inside}$$

where $\vec{E}_{inside}$ is the field in the hollow.





$$\vec{r} = \vec{b} + \vec{r}\,'$$

$$\vec{E}_{inside}(\vec{r}) = \vec{E}_R(\vec{r}) + \vec{E}_a(\vec{r})$$

el. field of large sphere

field of small sphere

Gauss's law: $$E_R(\vec{r}) \cdot 4\pi r^2 = \frac{1}{\varepsilon_0} \cdot \frac{4\pi}{3} r^3 \rho$$

$$\vec{E}_R(\vec{r}) = \frac{\rho\,\vec{r}}{3\varepsilon_0}$$

For small sphere $\vec{E}_a(\vec{r}\,') = \dfrac{(-\rho)\,\vec{r}\,'}{3\varepsilon_0}$

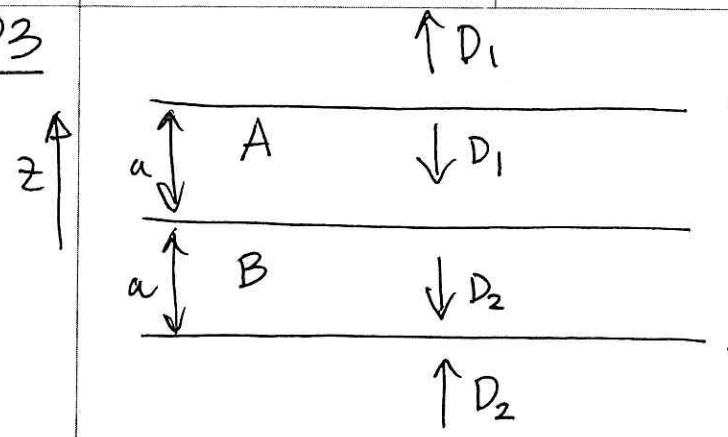
(it is negatively charged!)

$$\vec{E}_{inside}(\vec{r}) = \frac{\rho\,\vec{r}}{3\varepsilon_0} - \frac{\rho\,\vec{r}\,'}{3\varepsilon_0} = \frac{\rho}{3\varepsilon_0}\underbrace{(\vec{r} - \vec{r}\,')}_{\vec{b}} = \frac{\rho\,\vec{b}}{3\varepsilon_0}$$

Field inside the hollow cavity is constant!

$$\vec{F}_q = \frac{q\rho\,\vec{b}}{3\varepsilon_0} = \frac{q\rho\,b}{3\varepsilon_0}\,\hat{y}.$$

P3



$\varepsilon_{r,1} = 2 \Rightarrow \varepsilon_1 = 2\varepsilon_0$

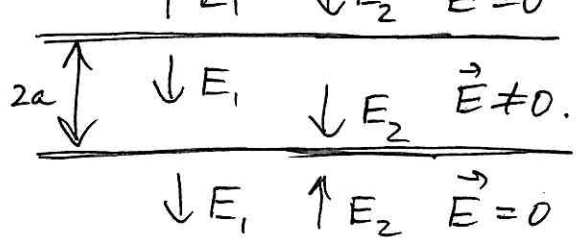
$\varepsilon_{r,2} = 1.5 \Rightarrow \varepsilon_2 = 1.5\varepsilon_0$

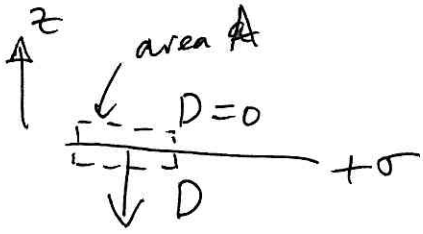
Plate $(+\sigma)$ creates $\vec{D}_1$; plate $(-\sigma)$ creates $\vec{D}_2$.

$$\vec{D} = \vec{D}_1 + \vec{D}_2 .$$

Outside the capacitor $\vec{D}_1 = \varepsilon_0 \vec{E}_1$, $\vec{D}_2 = \varepsilon_0 \vec{E}_2$ and

$$\vec{D} = \varepsilon_0 \vec{E} = \varepsilon_0 (\vec{E}_1 + \vec{E}_2) = 0$$





Slab A

$$D_A A = Q_{f,enc} = (+\sigma) \cdot A \Rightarrow D_A = \sigma$$

$$\vec{D}_A = \sigma(-\hat{z})$$

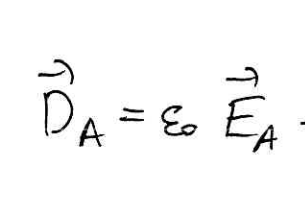
$$\vec{D}_A = \varepsilon \vec{E}_A \Rightarrow \vec{E}_A = \frac{1}{\varepsilon_1} \vec{D}_A = \frac{\sigma(-\hat{z})}{2\varepsilon_0}$$

el. field inside slab A.

by def: $\vec{D}_A = \varepsilon_0 \vec{E}_A + \vec{P}_A \Rightarrow \vec{P}_A = \vec{D}_A - \varepsilon_0 \vec{E}_A = \sigma(-\hat{z}) - \varepsilon_0 \frac{\sigma(-\hat{z})}{2\varepsilon_0}$

$$= \frac{1}{2}\sigma(-\hat{z}).$$

Slab B



$$D_B \cdot \text{Area} = (-\sigma)\ \text{Area}$$

$$\vec{D}_B = (-\sigma)(+\hat{z}) = \sigma(-\hat{z}).$$

$$\vec{E}_B = \frac{1}{\varepsilon_2}\vec{D}_B = \frac{-\sigma\hat{z}}{1.5\varepsilon_0} = \frac{-2\sigma\hat{z}}{3\varepsilon_0}$$

$$\vec{P}_B = \vec{D}_B - \varepsilon_0\vec{E}_B = -\sigma\hat{z} - \varepsilon_0\left(\frac{-2\sigma}{3\varepsilon_0}\hat{z}\right) = -\frac{1}{3}\sigma\hat{z}.$$

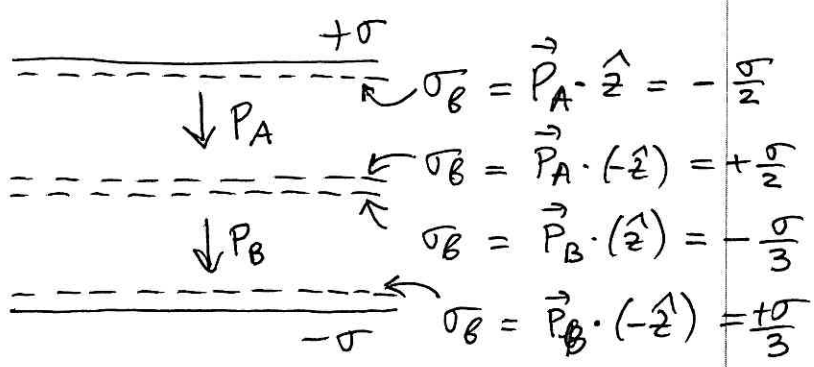
Thus (a) $\vec{D}_A = -\sigma\hat{z}$, $\vec{D}_B = -\sigma\hat{z}$

(b) $\vec{E}_A = -\frac{\sigma}{2\varepsilon_0}\hat{z}$, $\vec{E}_B = -\frac{2\sigma}{3\varepsilon_0}\hat{z}$

(c) $\vec{P}_A = -\frac{\sigma}{2}\hat{z}$, $\vec{P}_B = -\frac{\sigma}{3}\hat{z}$.

(d) $V = -\int \vec{E}\,d\vec{\ell} = -E_A\cdot a - E_B\cdot a =$

$$= \frac{\sigma a}{2\varepsilon_0} + \frac{2\sigma a}{3\varepsilon_0} = \frac{7\sigma a}{6\varepsilon_0}$$

(e) $\rho_b = -\vec{\nabla}\cdot\vec{P} = 0$ everywhere as polarization is constant inside slabs A and B

$\sigma_b = \vec{P}\cdot\hat{n}$